MIL-G-19807 (AER)
14 January 1957

MILITARY SPECIFICATION

GAGE, AIRCRAFT, CAPACITANCE, LIQUID OXYGEN QUANTITY INDICATOR REPEATERS, TYPES MC-5R, MC-6R, MC-7R, MC-8R

This specification has been approved by the
Bureau of Aeronautics, Department of the Navy

1. SCOPE

1.1 This specification covers a series of capacitance type liquid oxygen quantity gage system indicator repeaters. Type MC-8R is used with 2 1/2 liter primary indicators, MC-5R with 5 liter primary indicators, MC-6R with 10 liter primary indicators and MC-7R with 25 liter primary indicators.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings, and publication of the issue in effect on date of invitation for bids, form part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

TT-C-595	Colors; (for) Ready-Mixed Paints

Military

MIL-C-9082(ASG)	Converter, Oxygen, Aircraft, Liquid to Gaseous, General Specification For
MIL-C-19328(AER)	Converter, Oxygen, Aircraft, Liquid to Gaseous, Type MB-5 (5 liter)
MIL-C-19801(AER)	Converter, Oxygen, Aircraft, Liquid to Gaseous, Type MB-8 (2 1/2 liter)
MIL-C-19803(AER)	Converter, Oxygen, Aircraft, Liquid to Gaseous, Type MB-6 (10 liter)
MIL-C-19802(AER)	Converter, Oxygen, Aircraft, Liquid to Gaseous, Type MB-7 (25 liter)
MIL-G-19053(AER)	Gage, Aircraft, Capacitance, Liquid Oxygen Converter, General Specification for
MIL-G-19327(AER)	Gage, Aircraft, Capacitance, Liquid Oxygen Converter, Type MC-5 (5 liter)
MIL-G-19806(AER)	Gage, Aircraft, Capacitance, Liquid Oxygen Converter, Type MC-8 (2 1/2 liter)
MIL-G-19804(AER)	Gage, Aircraft, Capacitance, Liquid Oxygen Converter, Type MC-6 (10 liter)
MIL-G-19805(AER)	Gage, Aircraft, Capacitance, Liquid Oxygen Converter, Type MC-7 (25 liter)
MIL-I-6181	Interference Limits and Tests; Aircraft, Electrical and Electronic Equipment
MIL-P-6064	Packaging of light weight aircraft accessories
MIL-W-16878	Wire, Electrical (Insulated, high temperature)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T06:42Z
Check the source to verify that this is the current version before use.

DRAWINGS

Air Force-Navy Aeronautical

AND-10404	Pointers-Standard Design of Aircraft Instrument
AND-10412	Case-2-inch size instrument, Standard dimensions for

Military

MS-33558	Numerals and Letters, Aircraft Instrument Dial

(When requesting specifications, standards, drawings, and publications, refer to both title and number. Copies of this specification and applicable specifications may be obtained upon application to the Commanding Officer, Naval Aviation Supply Depot, Philadelphia 11, Pennsylvania, Attention: Code ODPT.)

3. REQUIREMENTS

3.1 General - The requirements in MIL-G-19053, MIL-C-9082 and the appropriate detail gage and converter specifications are applicable as requirements of this specification. The primary indicator in this system shall be in accordance with the applicable detail specification (such as MIL-G-19327 for the 5 liter gage.

3.1.1 In the event that the requirements of this specification and any applicable specifications conflict, the requirements of this specification shall govern.

3.2 Design - The indicator repeater defined herein, when used with the Primary indicator, shall constitute a dual indication gage system.

3.2.1 Case - The indicator repeater case shall be in accordance with Drawing AND-10412 and as shown in figure I.

3.2.2 Indicator Repeater - The indicator repeater shall obtain its input signal from the Primary indicator and shall be hermetically sealed. In a system, Repeater indication shall not differ from Primary indication by more than 2.0 percent of full scale plus 1.0 percent of gage indication.

3.2.3 Dial - The indicator repeater dial shall conform to Figure 1 of the Primary indicator detail specification.

3.2.3.1 Dial Markings - The dial markings shall conform to Figure 1 of the Primary indicator detail specification.

3.2.3.2 Pointer - The pointer shall conform to Figure 1 of the Primary indicator detail specification.

3.2.4 Low Level Warning - The indicator repeater shall not incorporate a low-level warning circuit. Low level warning lights associated with this indicator shall be controlled by the Primary indicator.

3.2.5 Power Failure - Power failure indication presentation shall conform to Figure 1 of the Primary indicator detail specification.

3.2.6 Field Adjustments - The indicator repeater shall be designed to operate without field adjustments. However, factory adjustment provisions shall be incorporated.

3.2.7 Fogging - The indicator repeater shall be kept at a temperature of -65°F ± 2° for at least 4 hours. There shall be no evidence of fogging of the cover glass during this period.

3.2.8 Weight - The weight of the indicator repeater shall not exceed 1.5 pounds.

3.2.9 Electrical Connector - Electrical connector(s) shall be in accordance with Figure 2 of this specification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T06:42Z
Check the source to verify that this is the current version before use.

3.2.10 Input Power - Indicator repeater input power requirements shall not exceed 15 volt-amperes not including integral lighting requirements.

3.2.11 Press to Test Switch - The indicator repeater shall not incorporate a test switch circuit. The test switch associated with this indicator shall be electrically paralleled with the Primary indicator test switch.

3.3 Identification of Product

3.3.1 Nameplate - A nameplate shall be securely attached to each separately mounted unit and shall be legibly and permanently marked with the following information:

Indicator Repeater
GAGE, LIQUID OXYGEN QUANTITY
Type MC-5R, MC-6R, MC-7R, MC-8R (as applicable, see Para. 1.1)
115 volts, 400 cycle, Single Phase
Manufacturer's Part No.
Manufacturer's Serial No.
Stock No. (Navy)
Contract or Order No.
Manufacturer's Name or Trade-mark
U.S.

3.4 Performance - The indicator repeater shall be capable of meeting the performance requirements of section 4.

4. QUALITY ASSURANCE PROVISIONS

4.1 General - The methods of sampling, inspection and tests shall be in accordance with MIL-G-19053 and as specified herein.

4.2 Test Instruments - Unless otherwise specified, the indicator repeater shall be connected to a master test instrument.

4.2.1 Master Test Instrument - When used, the master test instrument shall be capable of simulating the nominal Primary indicator repeater signal within 0.2 percent of full scale indication. The output shall be continuous and calibrated from 0 to 100 percent. Zero percent shall be the equivalent of empty and 100 percent shall be the equivalent of full. In all subsequent tests and measurements specified herein, the tolerance of the measuring equipment shall be added to the specified tolerance for the given test or measurement.

4.3 Test Methods - Testing shall be in accordance with applicable portions of MIL-G-19053 and as specified herein.

4.3.1 Examination of Product - Each unit shall be examined to determine conformance to the general specification and this specification with respect to materials, workmanship, weight and marking.

4.3.2 Indicator Operation at Room Temperature - The indicator repeater shall be operated with a master test instrument in lieu of a Primary indicator. The repeater pointer shall be set at the center of each major dial graduation by adjusting the master test instrument. The master test instrument readings shall be noted and shall be within 1.0 percent of full scale indication. These values shall be used as room temperature reference readings in later tests. (For separate indicators and amplifiers, the tolerance shall be 1.5 percent of full scale indication.)

4.3.3 Sealing - The hermetic seal of the units shall be tested in accordance with the general specification.

4.3.4 Dielectric Test - Dielectric testing shall be in accordance with the Primary indicator detail specification and shall not be conducted between points where circuit elements would be damaged by the test voltage.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T06:42Z
Check the source to verify that this is the current version before use.

4.3.5 Power Failure - Interruption of the input power to the indicator repeater shall actuate the power failure warning device.

4.3.6 Low Level Warning - Not applicable to the indicator repeater.

4.3.7 Temperature Tests

4.3.7.1 High Temperature - At the end of the four-hour period during which time the power shall be applied and while at the temperature specified in the general specification, the indicator repeater shall be tested in accordance with the applicable portion of the general specification. The scale error readings at each major graduation shall not differ from the room temperature reference readings by more than 0.75 percent of full scale. (The master test instrument shall remain room temperature.)

4.3.7.1.1 Return to Room Temperature - After return to room temperature, the scale error readings at each major graduation shall not differ from the room temperature reference readings by more than 0.5 percent of full scale.

4.3.7.2 Low Temperature - At the end of the four-hour period and at the temperature specified in the general specification, the indicator repeater shall be tested in accordance with the applicable portions of the general specification. The scale error readings at each major graduation shall not differ from the room temperature reference readings by more than 1.0 percent of full scale.

4.3.7.2.1 Return to Room Temperature - After return to room temperature, the scale error readings at each major graduation shall not differ from the room temperature reference readings by more than 0.5 percent of full scale.

4.3.8 Vibration Test - The indicator repeater shall be subjected to the Vibration Tests specified for the associated Primary indicator.

4.3.9 Shock Test - The indicator repeater shall be subjected to the Shock Tests specified for the associated Primary indicator.

4.3.10 Acceleration Test - The indicator repeater shall be subjected to the Acceleration Test specified for the associated Primary indicator.

4.3.11 Radio Interference

4.3.11.1 Conducted Interference - Maximum conducted radio interference on the power input leads of the indicator repeater for frequencies between 0.15 and 150 megacycles, shall not exceed the limits specified in MIL-I-6181.

4.3.11.2 Radiated Interference - Maximum radiated interference voltage for frequencies between 0.15 and 150 megacycles shall not exceed 12 microvolts.

4.3.12 Temperature Exposure Tests

4.3.12.1 High Temperature Exposure - At the end of the 48 hour period and while at the temperature specified in the general specification the indicator repeater shall be tested in accordance with the applicable portion of the general specification. The scale error readings at each major graduation shall not differ from the room temperature reference readings by more than 0.75 percent of full scale.

4.3.12.1.1 Return to Room Temperature - After return to room temperature the scale error readings at each major graduation shall not differ from the room temperature reference readings by more than 0.5 percent of full scale.

4.3.12.2 Low Temperature Exposure - At the end of the 48 hour period and while at the temperature specified in the general specification the indicator repeater shall be tested in accordance with the applicable portion of the general specification. The scale error readings at each major graduation shall not differ from the room temperature reference readings by more than 1.0 percent of full scale.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T06:42Z
Check the source to verify that this is the current version before use.

4.3.12.2.1 Return to Room Temperature - After return to room temperature, the scale error readings at each major graduation shall not differ from the room temperature reference readings by more than 0.5 percent of full scale.

4.3.13 Humidity - The humidity test shall be conducted in accordance with the applicable portions of the general specification. Scale error readings obtained within one hour after the test shall not differ from the room temperature reference readings by more than 2.0 percent of full scale.

4.3.14 Salt Spray - The indicator repeater shall be subjected to the salt spray test in accordance with the general specification. Scale error readings at each major graduations shall not differ from the room temperature reference readings by more than 2.0 percent of full scale.

4.3.15 One indicator repeater from each lot of 1,000 or less shall be picked at random by the cognizant government inspector and submitted to the Bureau of Aeronautics for additional tests.

5. PREPARATION FOR DELIVERY

5.1 Application - The requirements of the general specification shall apply.

5.2 Marking and Labeling - Marking shall be in accordance with the requirements of Specification MIL-P-0064 and as specified herein.

5.2.1 Packages - Each interior package shall be marked with the following information.

Indicator Repeater
GAGE, LIQUID OXYGEN QUANTITY
Specification MIL-G-19807 (Aer)
Stock No. (Navy)
Manufacturer's Part No.
Manufacturer's Serial No.
Contract or Order No.
Date of Manufacture
Name of Manufacturer
Name of Contractor (If different from manufacturer)
"IF IN STORAGE AFTER (DATE)*, THE INDICATOR SHALL BE TESTED AND INSPECTED BEFORE USE."

NOTE * This date shall be 1 year from date of inspection and shall be inserted by the Inspector at the time of acceptance.

6. NOTES

6.1 Intended Use - The indicator repeaters covered by the specification are intended to repeat the liquid oxygen quantity indicated by the primary Indicator (as defined by the applicable detail specification). This indicator repeater(s) is not affected by "G" forces or converter system pressure. The system indicator repeater(s) may be located at a remote point(s) from the indicator, within the aircraft.

6.2 Ordering Data - Requisitions, contracts and orders should specify whether overseas shipment is desired.

6.3 Qualification - With respect to products requiring qualification, awards will be made only for such products as have, prior to the bid opening date, been tested and approved for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T06:42Z
Check the source to verify that this is the current version before use.

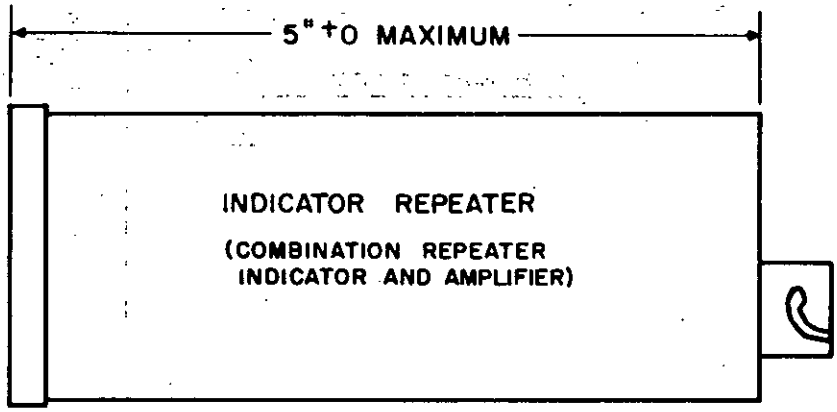
6.3.1 The attention of suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government, tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products covered by this specification may be obtained from the Chief, Bureau of Aeronautics, Department of the Navy, Washington 25, D.C.

6.3.2 It is to be understood that upon receipt of the Letter of Authorization, samples shall be furnished at no cost to the Government, and that the manufacturer shall pay the transportation charges to and from the designated point where tests are to be made. In the case of failure of the sample or samples submitted, consideration will be given to the request of the manufacturer for additional tests only after it has been clearly shown that changes have been made in the product which the Government considers sufficient to warrant additional tests.

6.4 Procuring Agency - In this specification, the term Procuring Agency refers to Bureau of Aeronautics, Department of the Navy.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the U. S. Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished or in any way supplied the said drawings, specifications, or other data is not to be inferred by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T06:42Z
Check the source to verify that this is the current version before use.



CASE IN ACCORDANCE WITH AND10412

FIGURE 1a PREFERRED CONFIGURATION

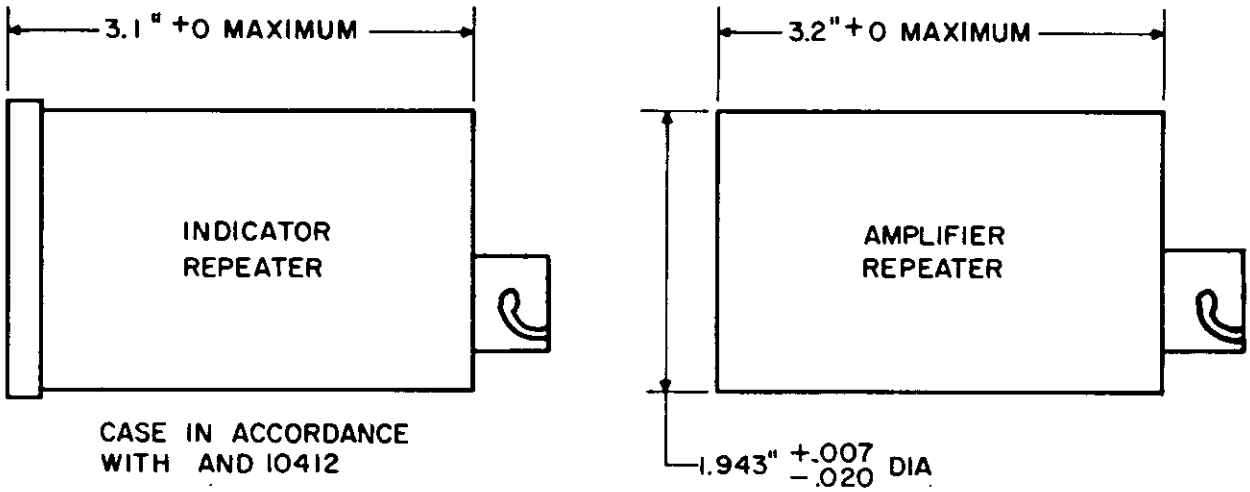


FIGURE 1b ALTERNATE CONFIGURATION

FIGURE 1
INDICATOR GAGE
LIQUID OXYGEN QUANTITY, REPEATER

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T06:42Z
Check the source to verify that this is the current version before use.

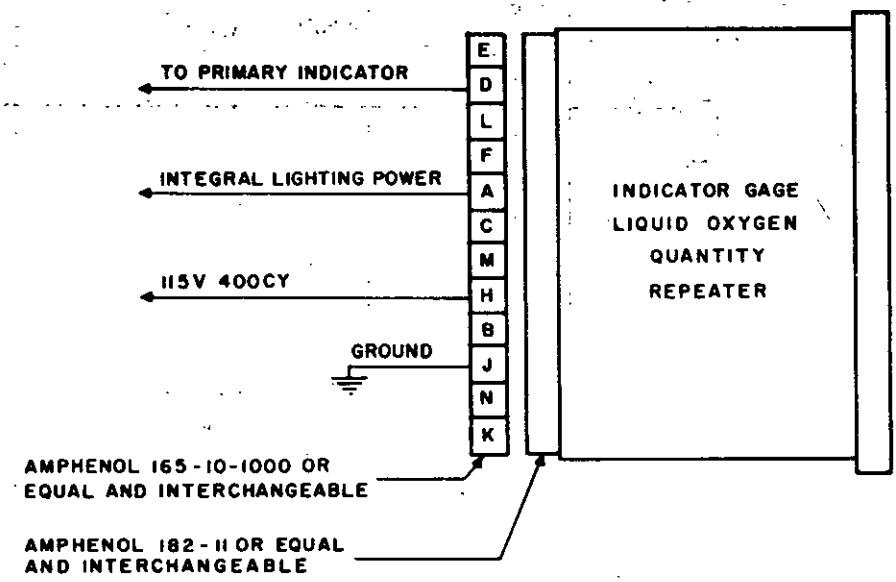


FIGURE 2a PREFERRED CONFIGURATION

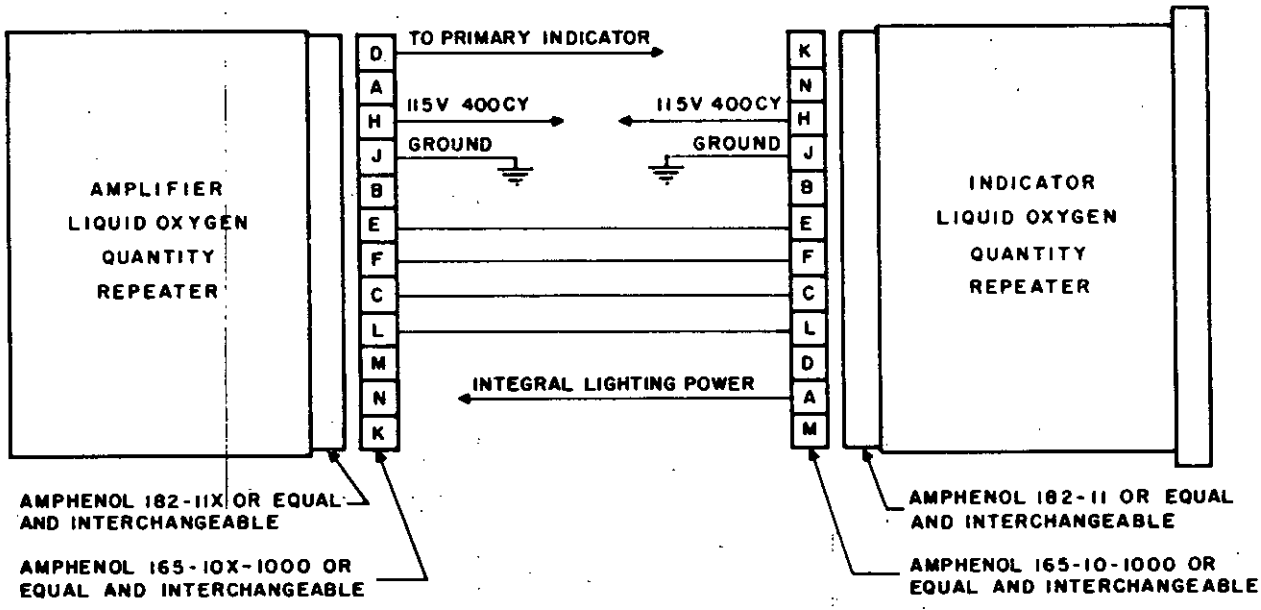


FIGURE 2b ALTERNATE CONFIGURATION

FIGURE 2.
GAGING SYSTEM ARRANGEMENT

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T06:42Z
Check the source to verify that this is the current version before use.